■ Oceanus Agentic RAG Platform

Oceanographic Data Intelligence & Multi-Agent RAG Architecture Overview

1. Executive Summary

Oceanus is an enterprise-grade AI platform designed for marine researchers, oceanographers, and climate scientists to explore, query, and visualize global **Argo Float ocean measurement datasets**. By replacing complex SQL and GIS workflows with natural language conversation, users can instantly analyze sea surface temperature anomalies, salinity profiles, and underwater pressure depth metrics across the world's oceans.

2. Key Features & Capabilities

- **Natural Language Ocean Querying:** Ask plain-English questions about float measurements, regional thermal patterns, or depth profiles.
- **Multi-Agent RAG Intelligence:** Coordinated suite of specialized LLM agents (Router, DB Tool Agent, Vector RAG Agent, Quality Evaluator).
- **Conversational Session Memory:** Remembers float IDs (e.g. *7902073*), regions discussed (*Arabian Sea*), and detail level preferences across multi-turn chats.
- **Interactive Geospatial Mapping:** Real-time Leaflet map displaying active float locations, pulsating markers, and depth trajectory popups.
- **Depth Profile Visualization:** Recharts graphics rendering Temperature (°C), Salinity (PSS), and Pressure (dbar) vs. depth and time.
- **Resilient Fallback Mode:** Operates with zero downtime by automatically loading local CSV datasets (88 floats, 460,000+ data rows) when offline.

3. Technical Architecture

Component	Technology / Framework	Role / Functionality
Frontend UI	Next.js 15, React 19, TailwindCSS	Interactive Dashboard, AI Chat Sidebar, Glassmorphism UI
Geospatial Map	Leaflet.js, Recharts, Three.js	Float cluster markers, trajectory popups, depth graphs
Backend Server	FastAPI, Uvicorn, Python 3.13	Unified REST API for Chat, Sessions, Metrics & Float Data
Multi-Agent Core	LangChain, Groq, OpenAI LLMs	Multi-cycle reasoning, quality evaluation, agent orchestration
Database / RAG	CockroachDB, Neo4j, Pinecone, CSV	Time-series measurements, graph relations, vector RAG

4. System Endpoints & Quick Start

- **Web Application Interface:** http://localhost:9002
- **Unified FastAPI Server:** http://localhost:8000
- **ReDoc Interactive API Specs:** http://localhost:8000/redoc
- **Float Map Data Stream:** http://localhost:8000/api/floats
- **System Health Check:** http://localhost:8000/health

Unified Start Command:

`python start_app.py` (or double click **start_oceanus.bat**)
